Experiment 9: Markov Model and Reinforcement Learning

Aim

To understand Markov models and implement basic reinforcement learning techniques using Python.

Exercises:

1. **Markov Model:** Define states and transition probabilities for a simple environment. Simulate state changes, visualize the results, and study the system's long-term behavior.
2. **Reward-Based Decision-Making:** Define actions and assign rewards or penalties. Observe how changing rewards affects the agent's decisions.
3. **Q-Learning:** Implement Q-learning to learn the best policy for the same environment. Visualize Q-values or the learned policy and plot a learning curve of cumulative rewards across training episodes.
4. **Exploration vs. Exploitation:** Try different epsilon values and observe how they affect the agent's learning.
5. **Discount Factor:** Change the discount factor γ and compare its effect on the learned policy.
6. **Policy Evaluation:** Compare different settings using cumulative rewards, steps taken, and success rate.